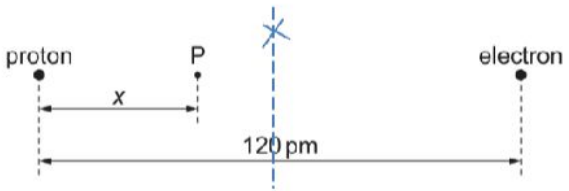
4(a)	work done per unit charge by an external force In bringing a small positive test charge from infinity to that point	[1] [1]
4(bi1)	potential (due to proton) = $\frac{1.6\times10^{-19}}{4\pi(8.85\times10^{-12})(10\times10^{-12})}$ = 143.86888 potential (due to electron) = $\frac{-1.6\times10^{-19}}{4\pi(8.85\times10^{-12})(110\times10^{-12})}$ = 13.07899 V = 143.86888 - 13.07899 = 130.7 ≈ 130 V (2sf)	[1] [1]
4(bi2)	$V_P = \frac{Q_{proton}}{4\pi\varepsilon_o x} + \frac{Q_{electron}}{4\pi\varepsilon_o(120\times10^{-12} - x)}$ $= \frac{1.6\times10^{-19}}{4\pi(8.85\times10^{-12})(30\times10^{-12})} + \frac{-1.6\times10^{-19}}{4\pi(8.85\times10^{-12})(90\times10^{-12})}$ = + 31.91 V = 32 V	[1]
4(bi3)	cross drawn midway, equidistant between the electron and the proton 	[1]

4(bi4)	 starting from (10, +130) to (110, –130) curve decreasing gradient until x = 60 pm, crossing V = 0 at (60, 0) and then getting steeper after x = 60 pm curve passing through (30, +32) and (90, -32)	[1] [1] [1]
4(bii)	$E_{midway} = \frac{Q_{proton}}{4\pi\varepsilon_o x^2} + \frac{Q_{electron}}{4\pi\varepsilon_o x^2}$ $= \frac{1.6\times10^{-19}}{4\pi(8.85\times10^{-12})(60\times10^{-12})^2} + \frac{1.6\times10^{-19}}{4\pi(8.85\times10^{-12})(60\times10^{-12})^2}$ = 8.0 x 10^{11} N C^{-1} (to the right)	[1] [1]
4(biii)	Rotate clockwise Accelerate upwards	[1] [1]